PhoneBoxSound

Your PC sound. Everywhere.

Turn the phones you already own into synchronised wireless speakers, a media remote and an intercom for your Windows PC. Nothing to install on the phones: they just open a web page.

No cloud · No account · No app install · Local network only · Free for non-commercial use

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SEE IT

The dashboard and the phone app

Recorded from the real app running in demo mode (a pretend player and synthetic music, so nothing personal is shown).



Dashboard on the PC: what's playing, every phone, studio layout, audio and pairing.



The phone: speaker, remote, touchpad, intercom.

GUIDED TOUR

Every screen and every setting, explained

A walk through the whole app with a spoken voice, a marker for where each thing happens (on the PC, on the phone, in the browser) and numbered steps: starting it, connecting phones, speakers, media, controlling the PC, intercom, security, the browser extension, bedtime and each group of settings.

The tour is the file docs/tour.html (a page you can step through) and docs/media/tour.mp4 (a video with a spoken voice).

START

One double-click

start.bat does the whole thing

IF THIS IS MISSING	IT DOES THIS, ONCE
Python	Installs Python 3.13 for your user, with winget or straight from python.org
Python packages	Installs what requirements.txt lists, for your user only
The built web app	Builds it with a portable Node.js from nodejs.org, kept in tools ode (a downloaded release already has it)
Nothing	Starts straight away and opens the dashboard

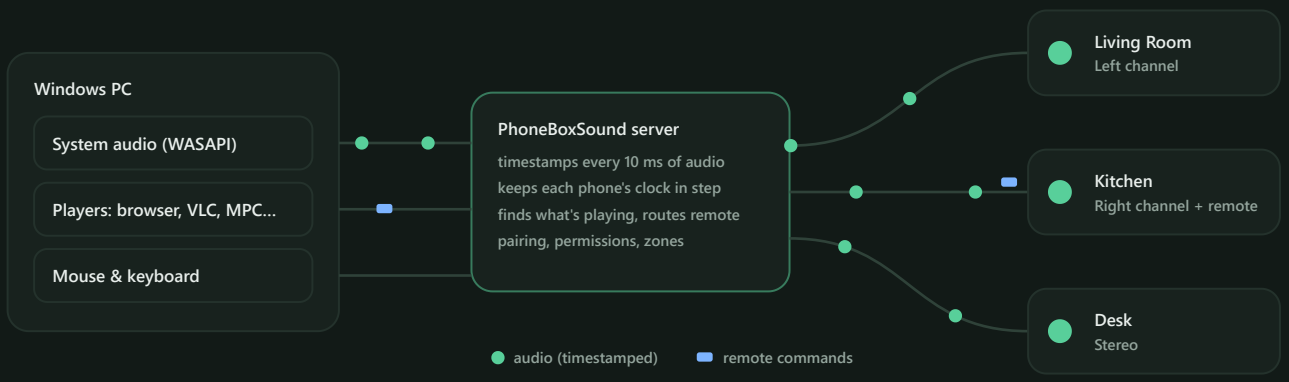
Before it downloads anything, the first start shows one screen: what it will download, from where, under which licence, that with winget it accepts winget's agreements for you, and who is responsible for what. Nothing happens until you type Y. `start.bat --check` only reports what is there and what is missing.

Then

STEP	WHERE
Open the dashboard	http://localhost:8080 on the PC
Connect a phone	Scan the QR code, or type the address and room code
Let it in	Approve it on the Devices page
Give it abilities	Remote page: one switch per phone and ability
Start with Windows	Settings, first switch

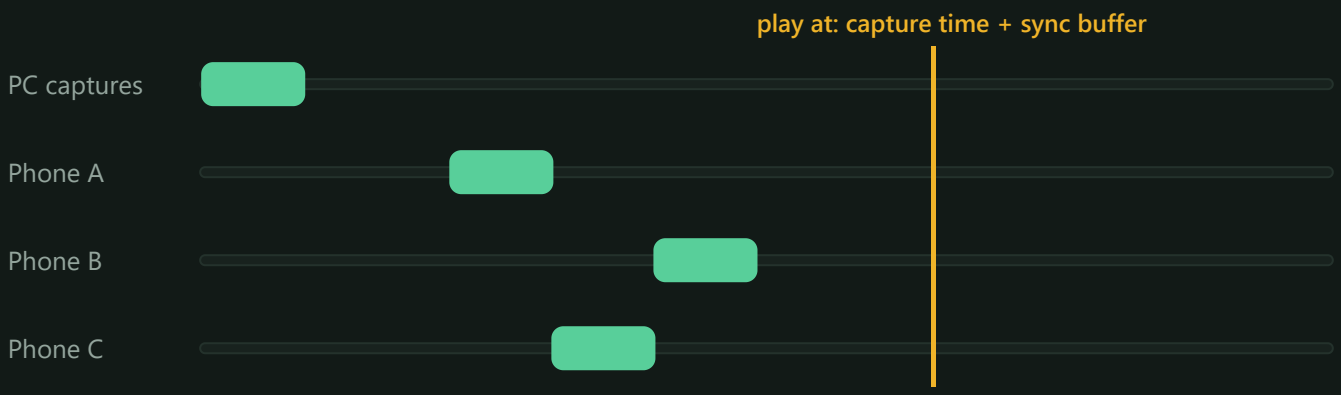
HOW IT WORKS

One capture, many phones, one moment



Why the phones stay together

Packets reach each phone at a different time: Wi-Fi is uneven. So a phone never plays audio "when it arrives". Every 10 ms block carries the moment it was captured, and every phone plays it at the **same agreed moment**: capture time + sync buffer.



Blocks arrive early, late, in between; all three turn white (play) on the same line.

The four mechanisms

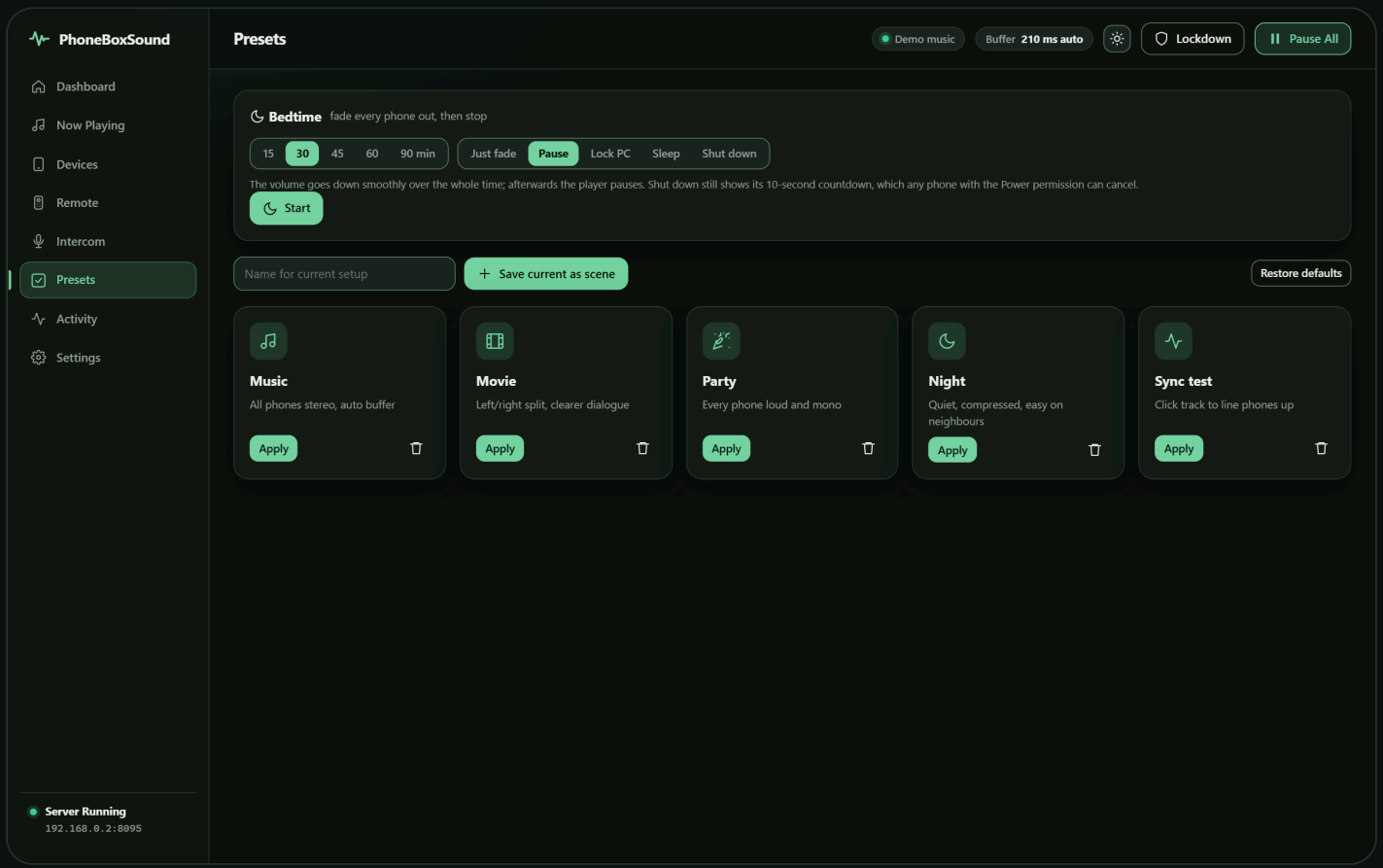
1. **Shared clock.** Each phone pings the PC and keeps only recent, fast round trips to learn the PC's clock. If new measurements disagree (the browser was asleep), old ones are thrown away at once.
2. **Scheduled playback.** Blocks are queued on the phone's audio hardware clock for an exact sample, not "as soon as possible".
3. **Drift correction.** Tiny errors are absorbed by playing up to 0.4% faster or slower: inaudible, no clicks.
4. **Adaptive buffer.** Each phone reports how much buffer its Wi-Fi really needs; the PC uses the largest, raises it instantly on trouble and lowers it gently.

ON THE PC

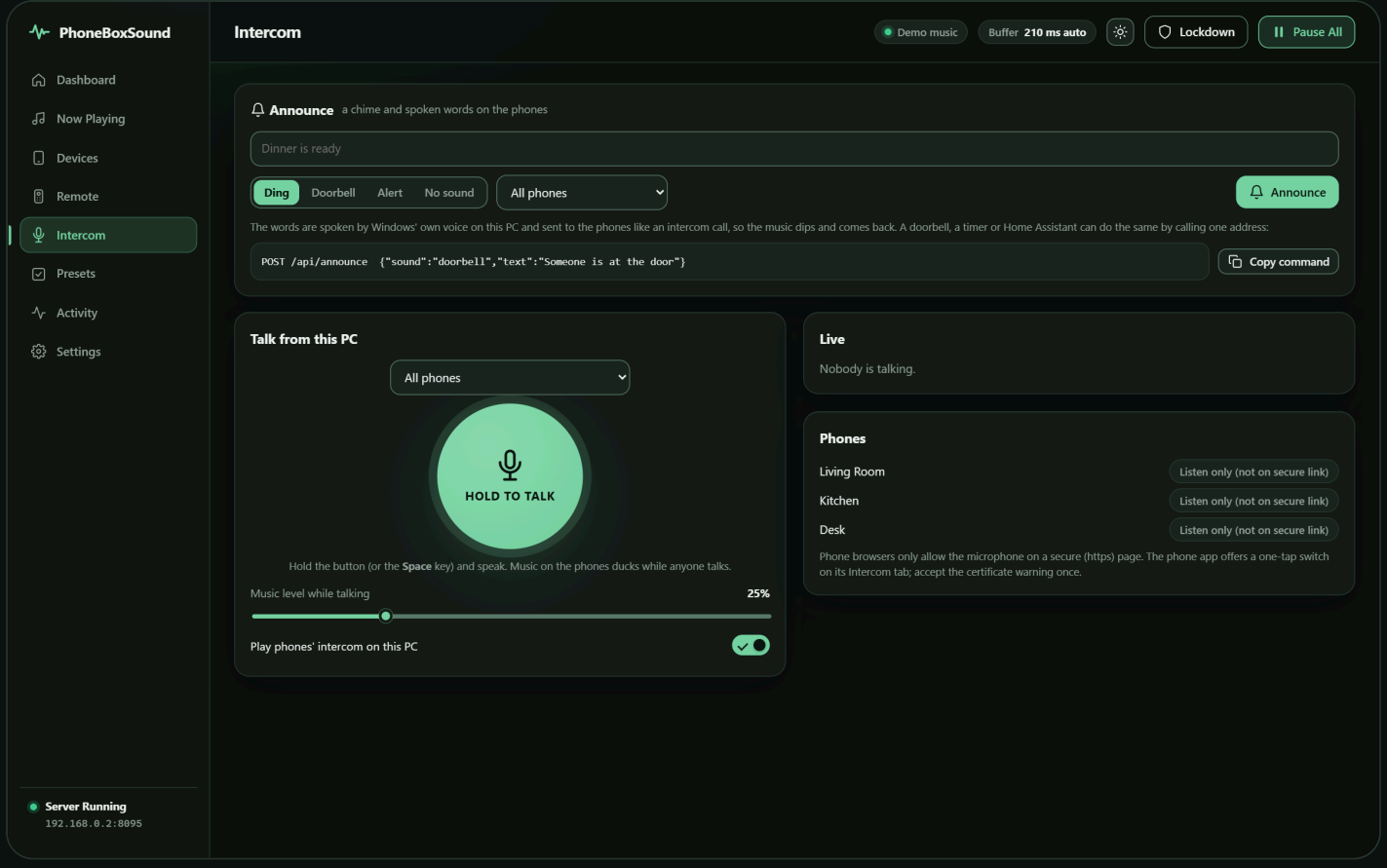
Dashboard at desktop scale



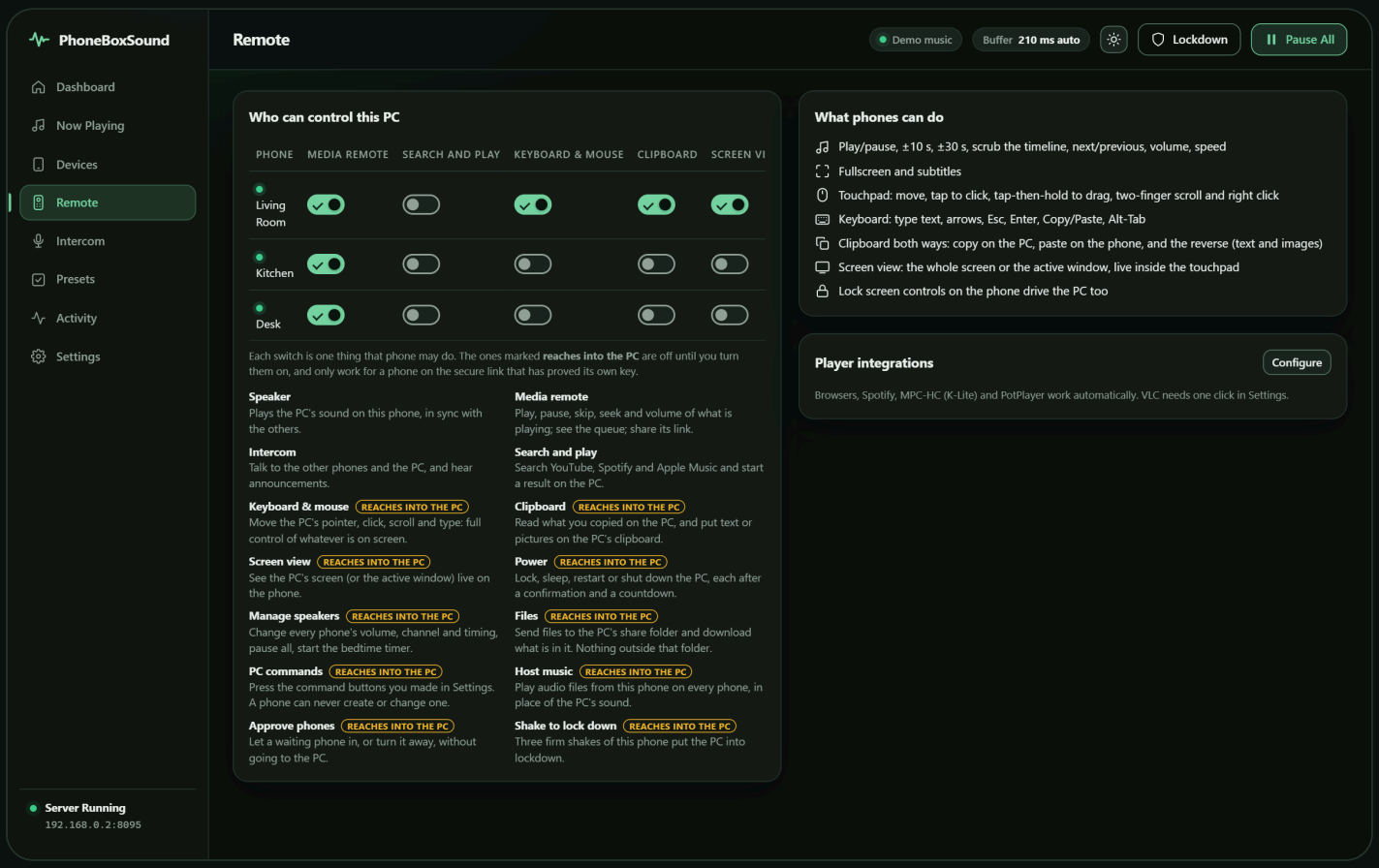
Now Playing. Follows whatever you started last and prefers the app actually making sound. Players without cover art show the live video frame.



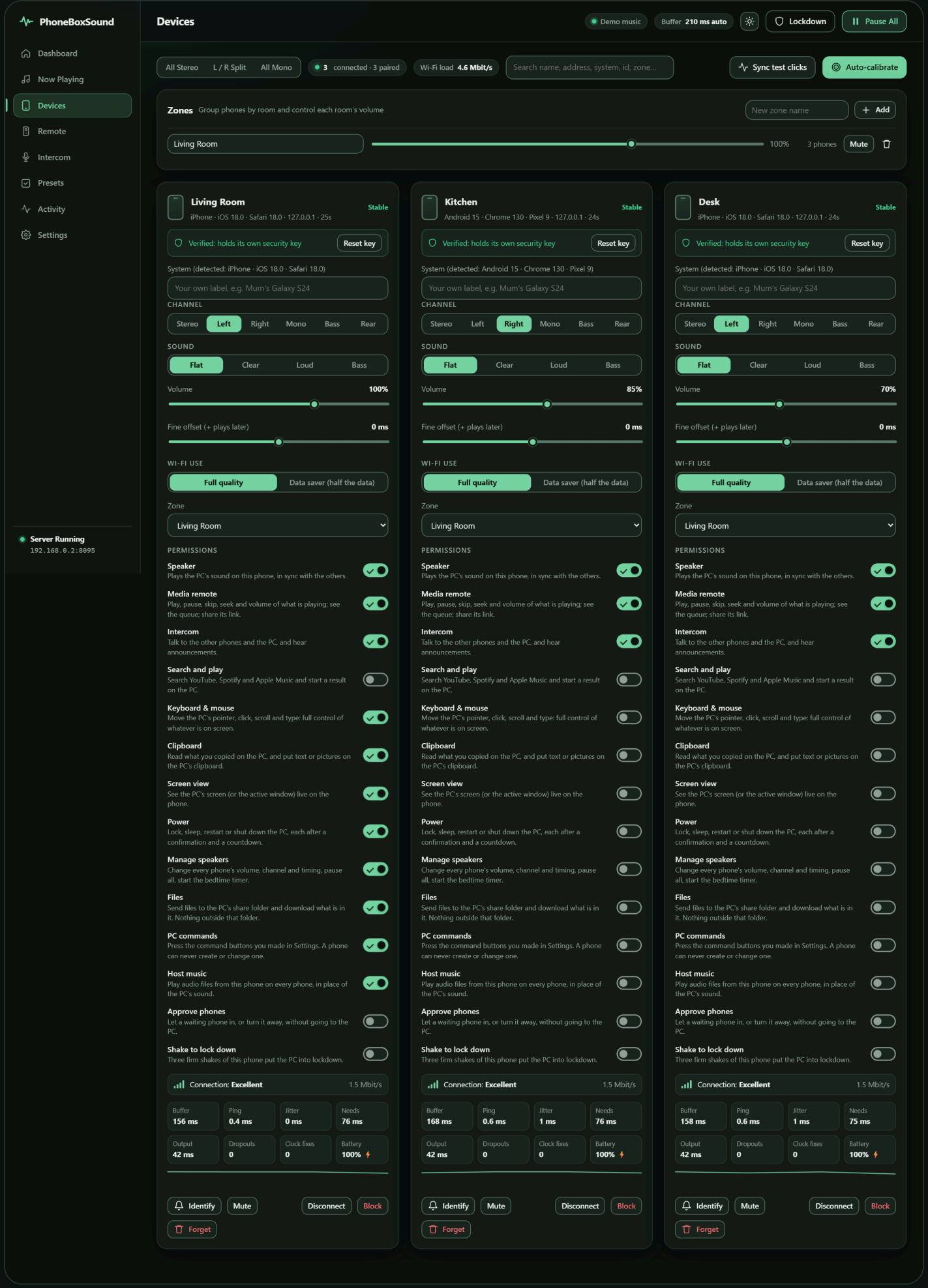
Scenes. Music, Movie, Party, Night, Sync test, or save your own setup.



Intercom. Hold to talk to everyone, a room or one phone. Music ducks while someone speaks.



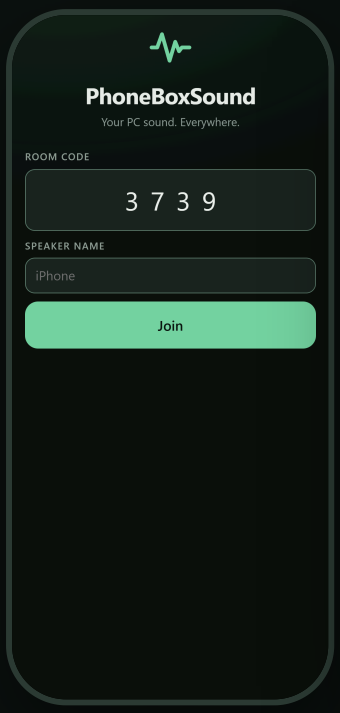
Permissions. Choose per phone: speaker, media remote, intercom, keyboard & mouse.



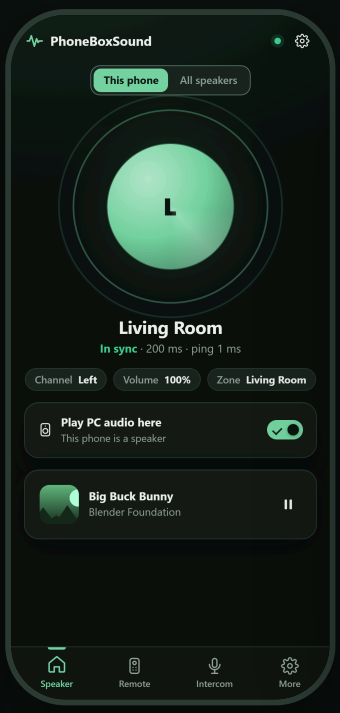
Devices. Zones, channel, sound profile, fine offset, live health for every phone, and one-click auto-calibration.

ON THE PHONE

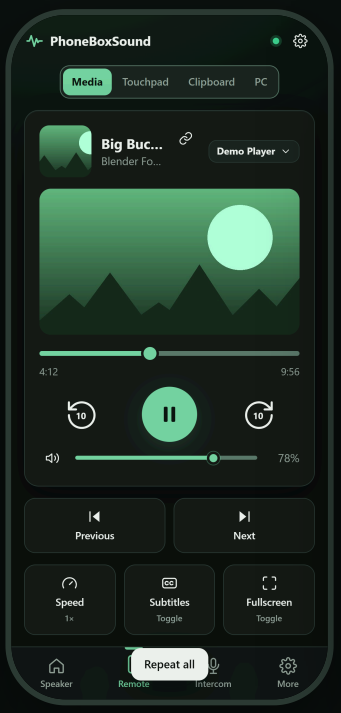
The phone app at phone scale



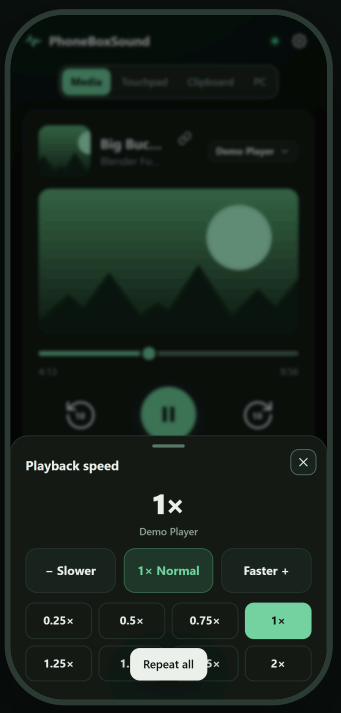
Join with a 4-digit code or a QR scan.



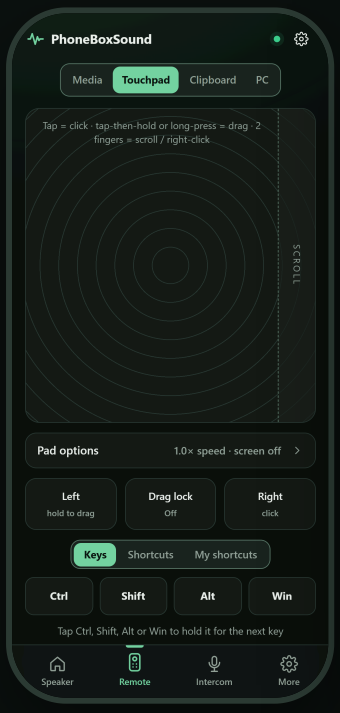
Speaker. Sync status at a glance.



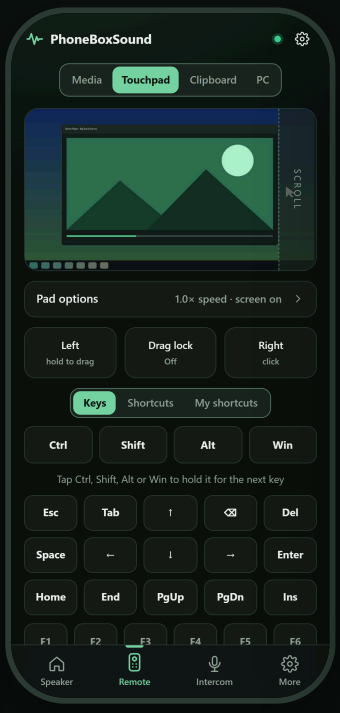
Remote. Drag the timeline to skim; swipe or double-tap the picture to skip.



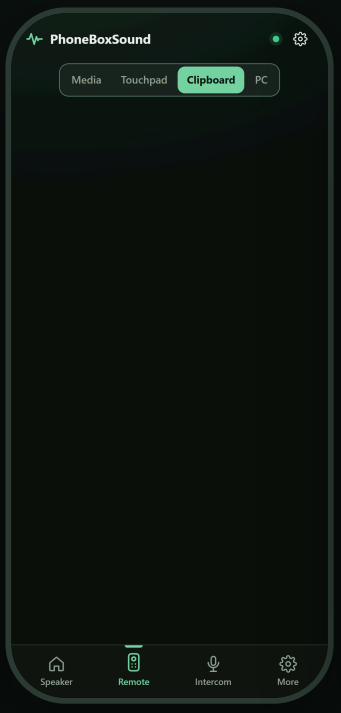
Speed. Slower, normal, faster, or exact presets.



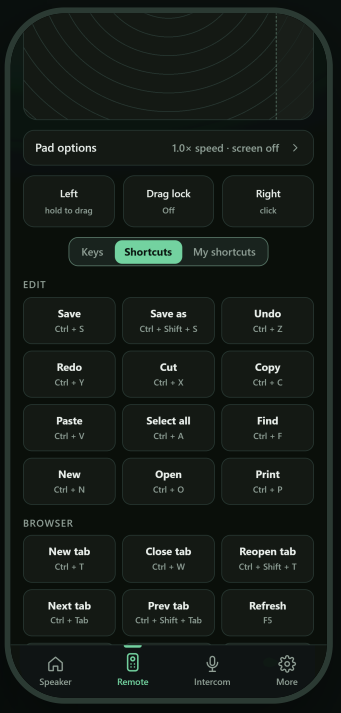
Touchpad. Tap, drag, two-finger scroll, drag lock, keys, typing.



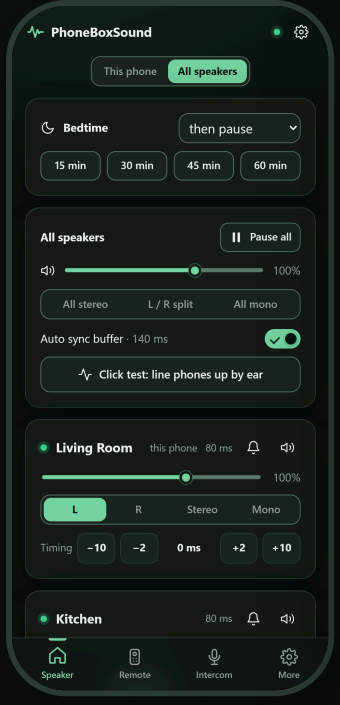
Screen view. The PC screen or the active window, live inside the touchpad.



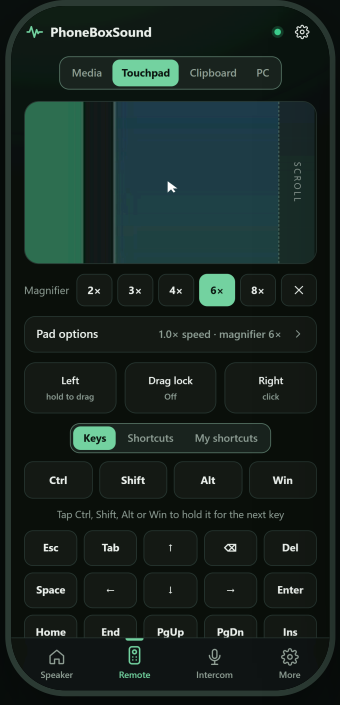
Clipboard. Copy on one, paste on the other: text and pictures, both ways.



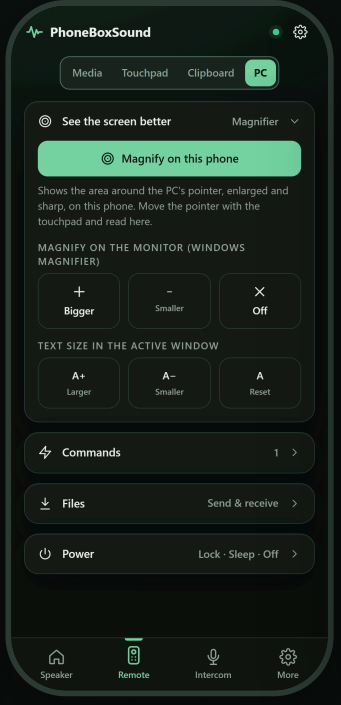
Shortcuts. Hold Ctrl / Shift / Alt / Win with any key, ready-made and personal shortcuts.



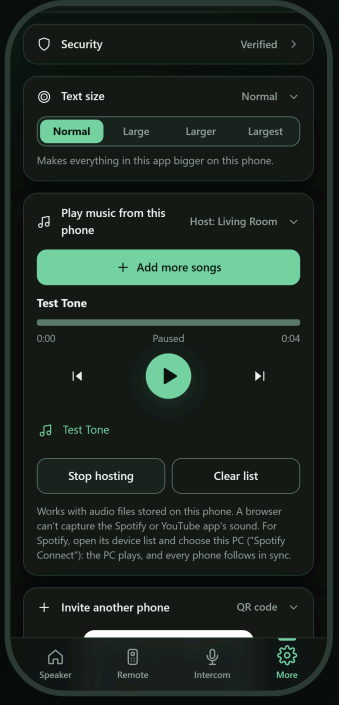
All speakers. Tune every phone's volume, channel and timing from one phone.



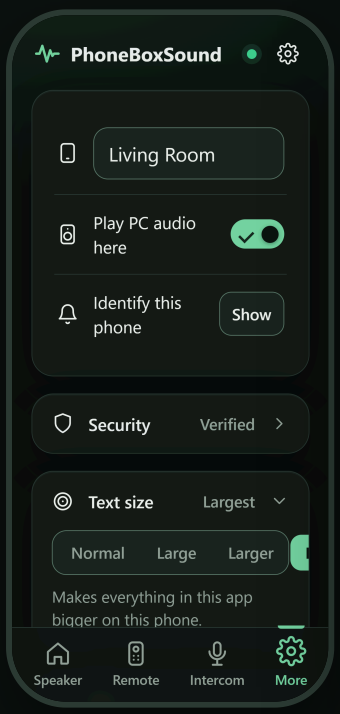
Magnifier. The area around the PC pointer, enlarged up to 8×, for reading small text.



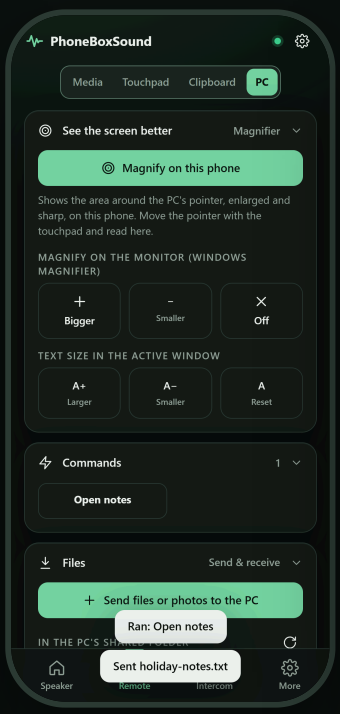
PC panel. See better, your own commands, files, power.



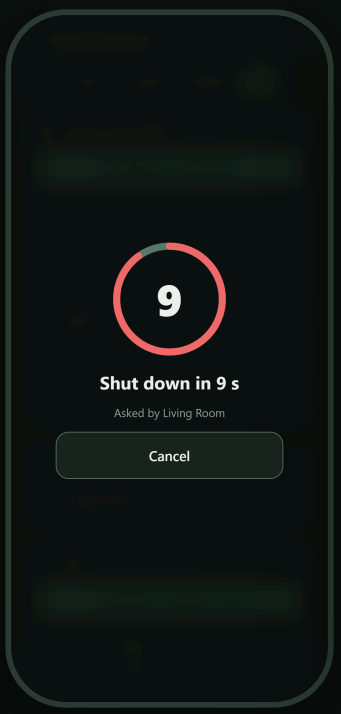
Host. Play songs stored on a phone on every phone in sync.



Text size. The whole app up to 1.5× larger.



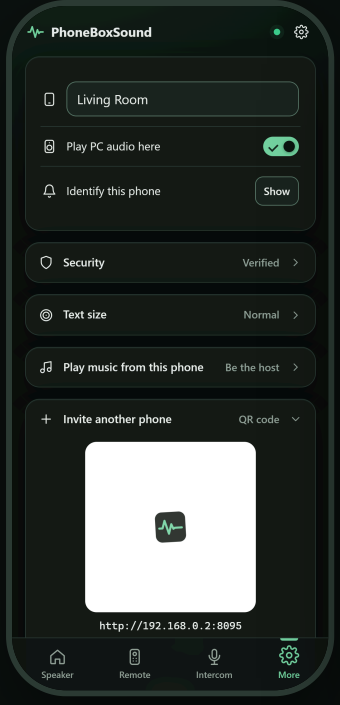
Power. Lock, sleep, restart, shut down: each asks first.



Countdown. Ten seconds to change your mind, on the phone or the PC.



Intercom. Hold, or tap to latch.



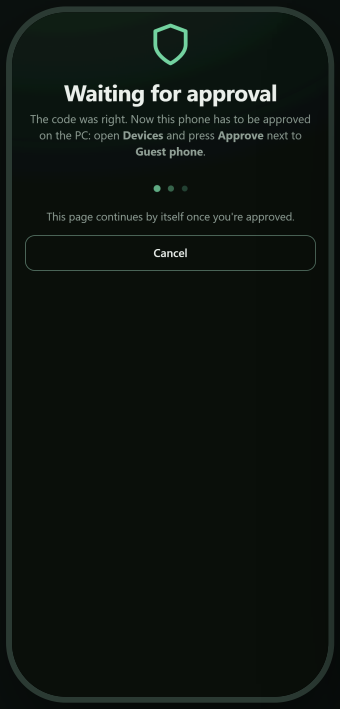
Invite. Any paired phone can show a QR for the next one.



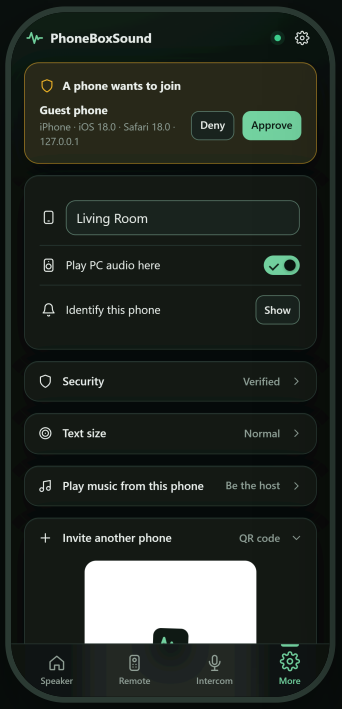
Themes. Five themes, dark or light.

SECURITY

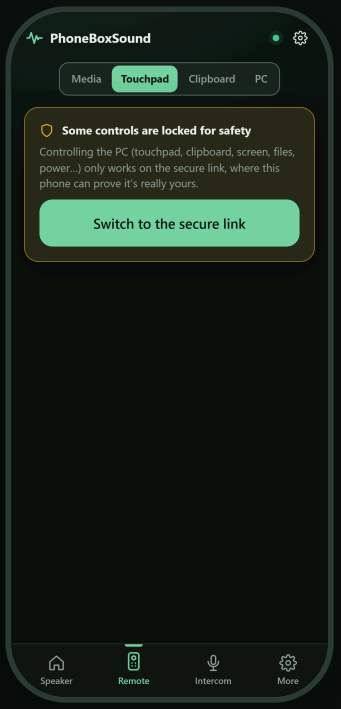
Who gets in, and what they can touch



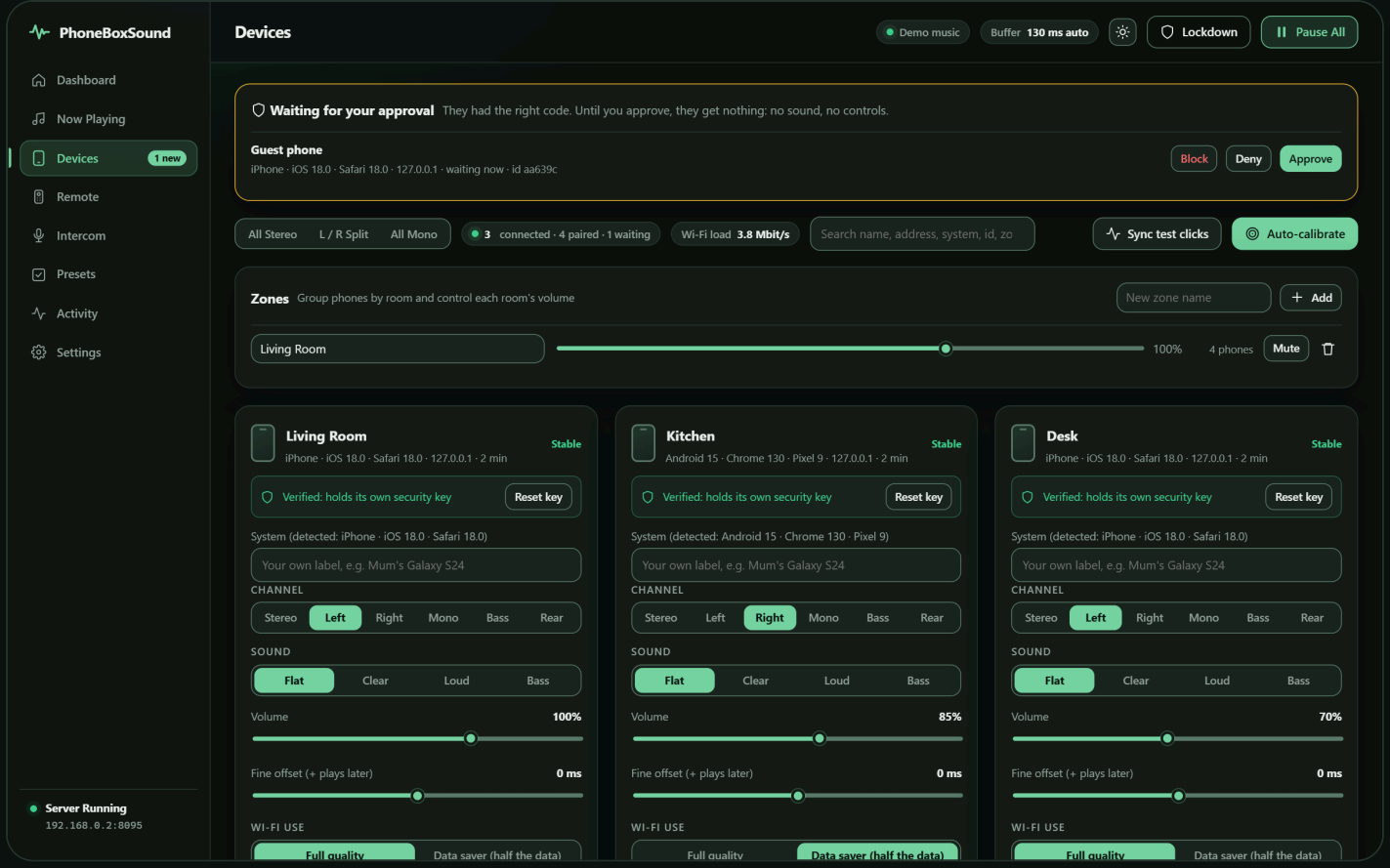
Waiting room. The right code only gets a new phone this far, until you approve it.



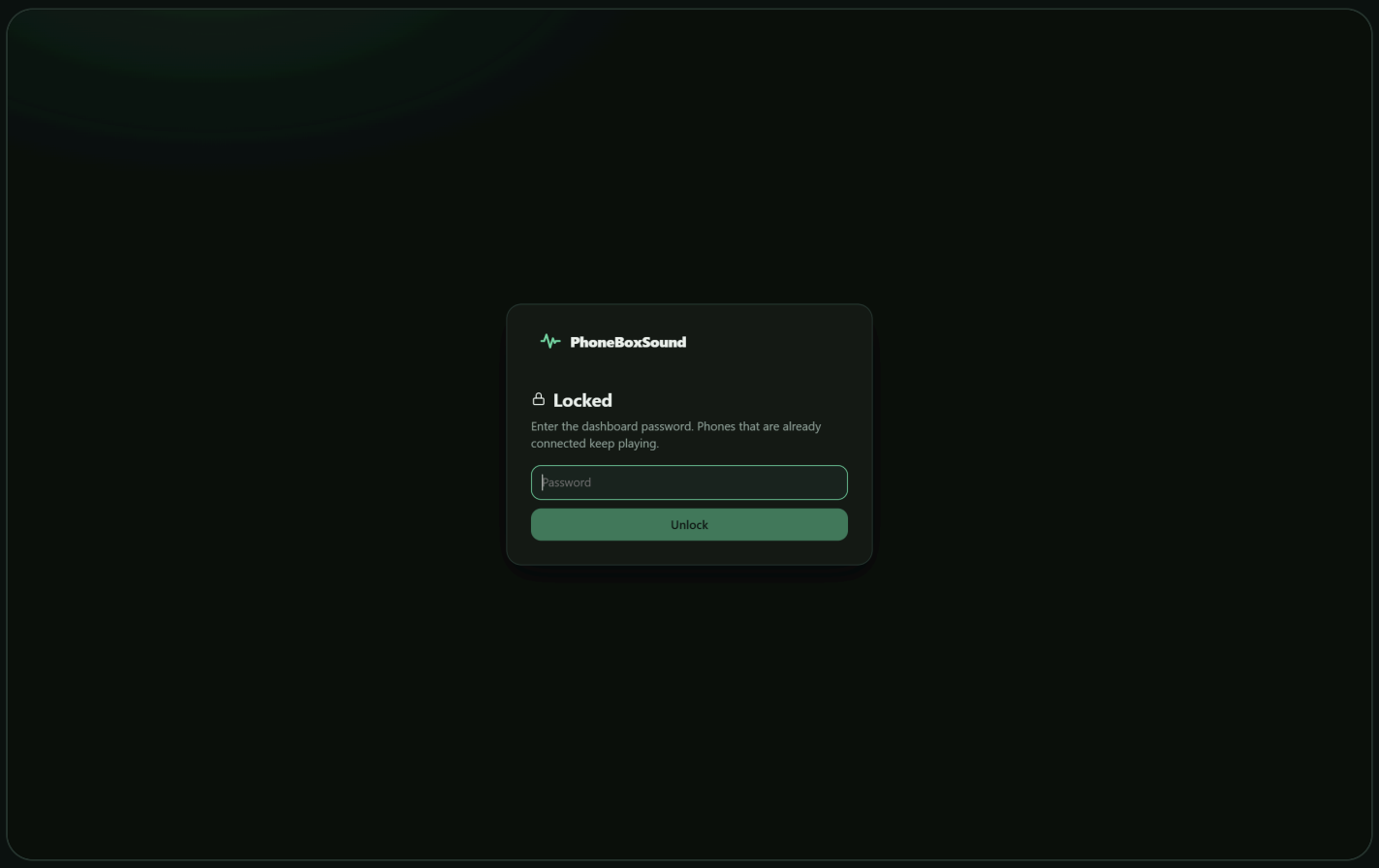
Approve from bed. A phone you trust with that permission can let others in.



Locked controls. PC control only on the secure link, for a phone that proved its own key.



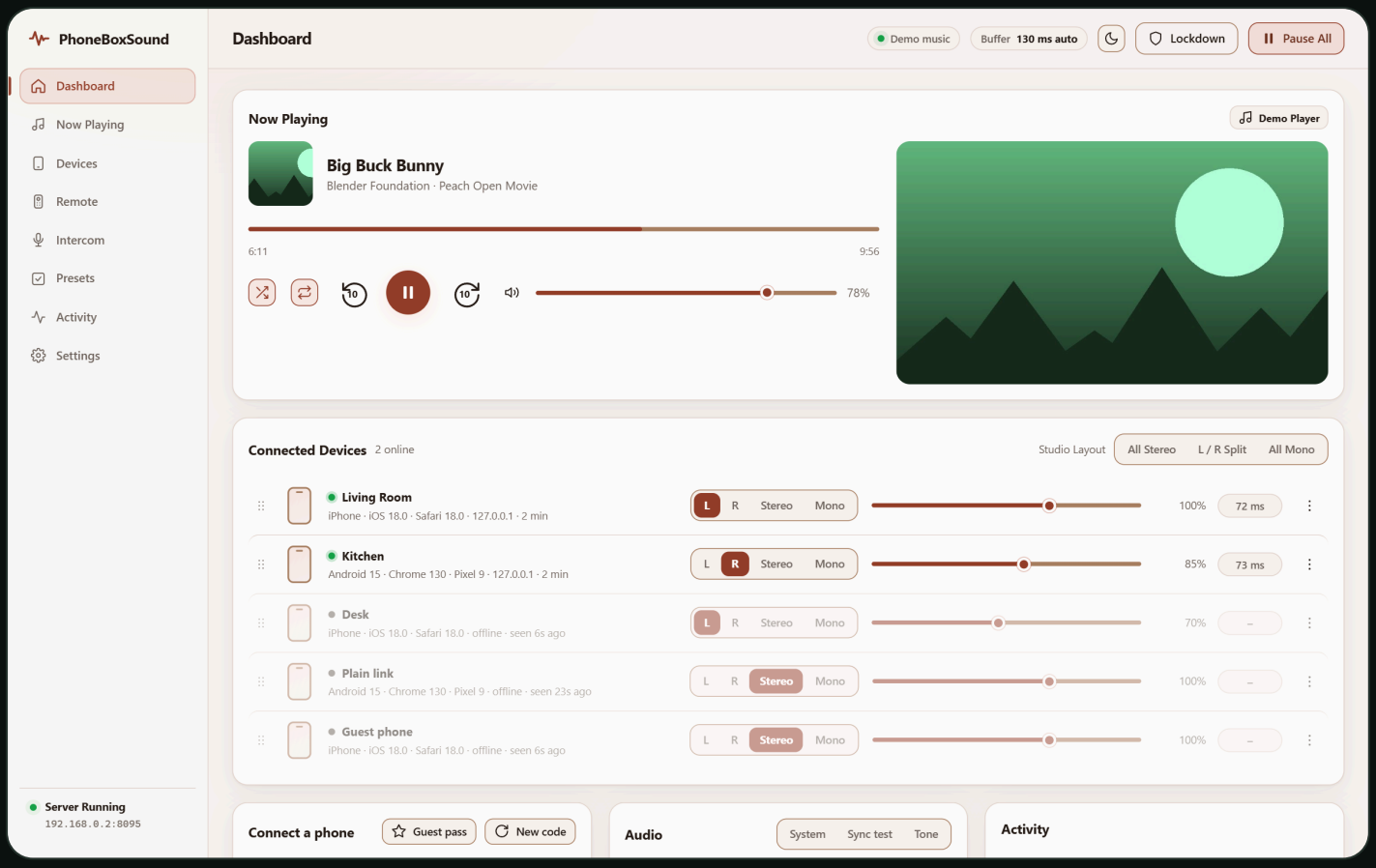
Approve, deny or block on the Devices page; search by name, address, system or id.



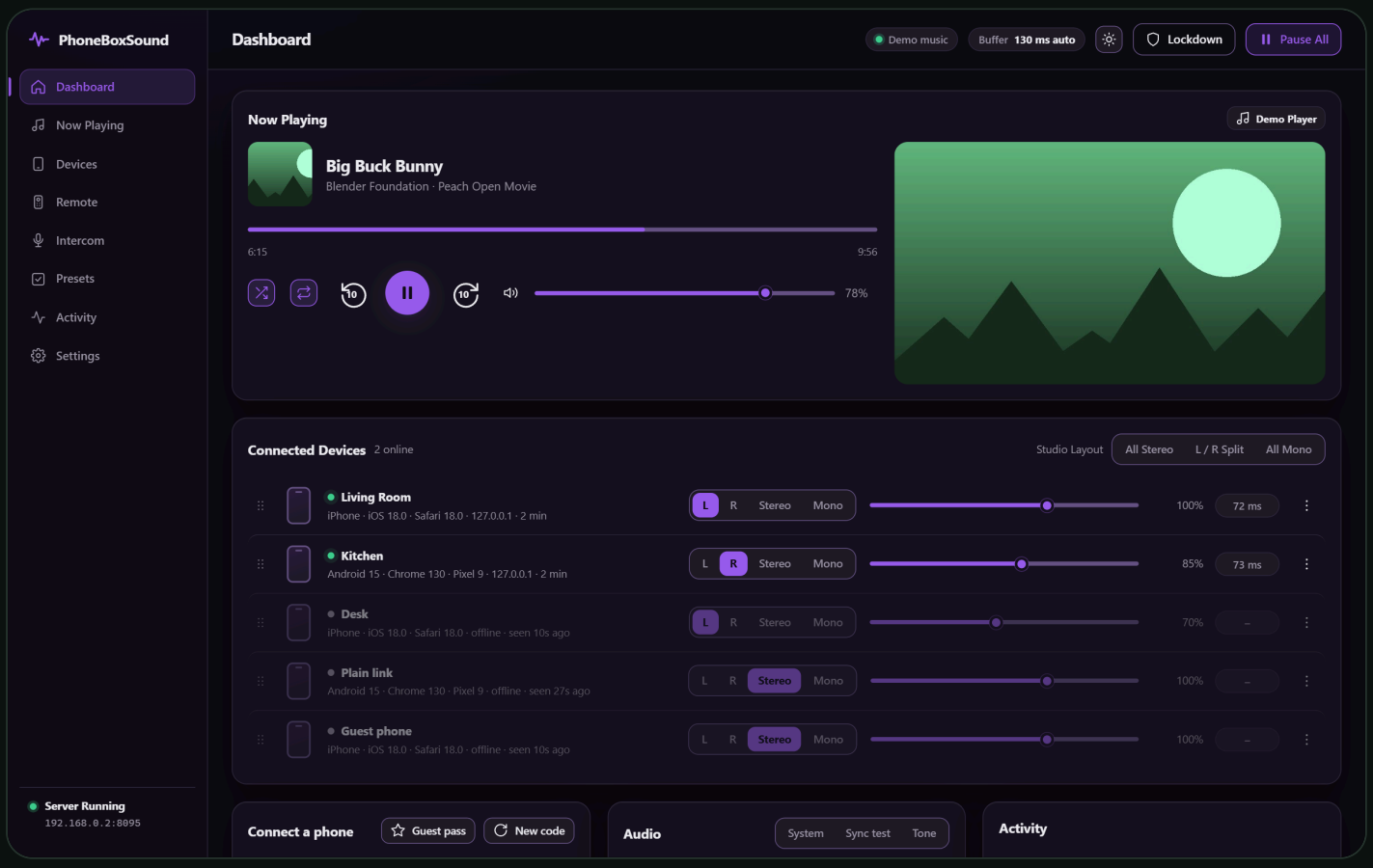
The dashboard behind a password; the storage behind it is encrypted.

THEMES

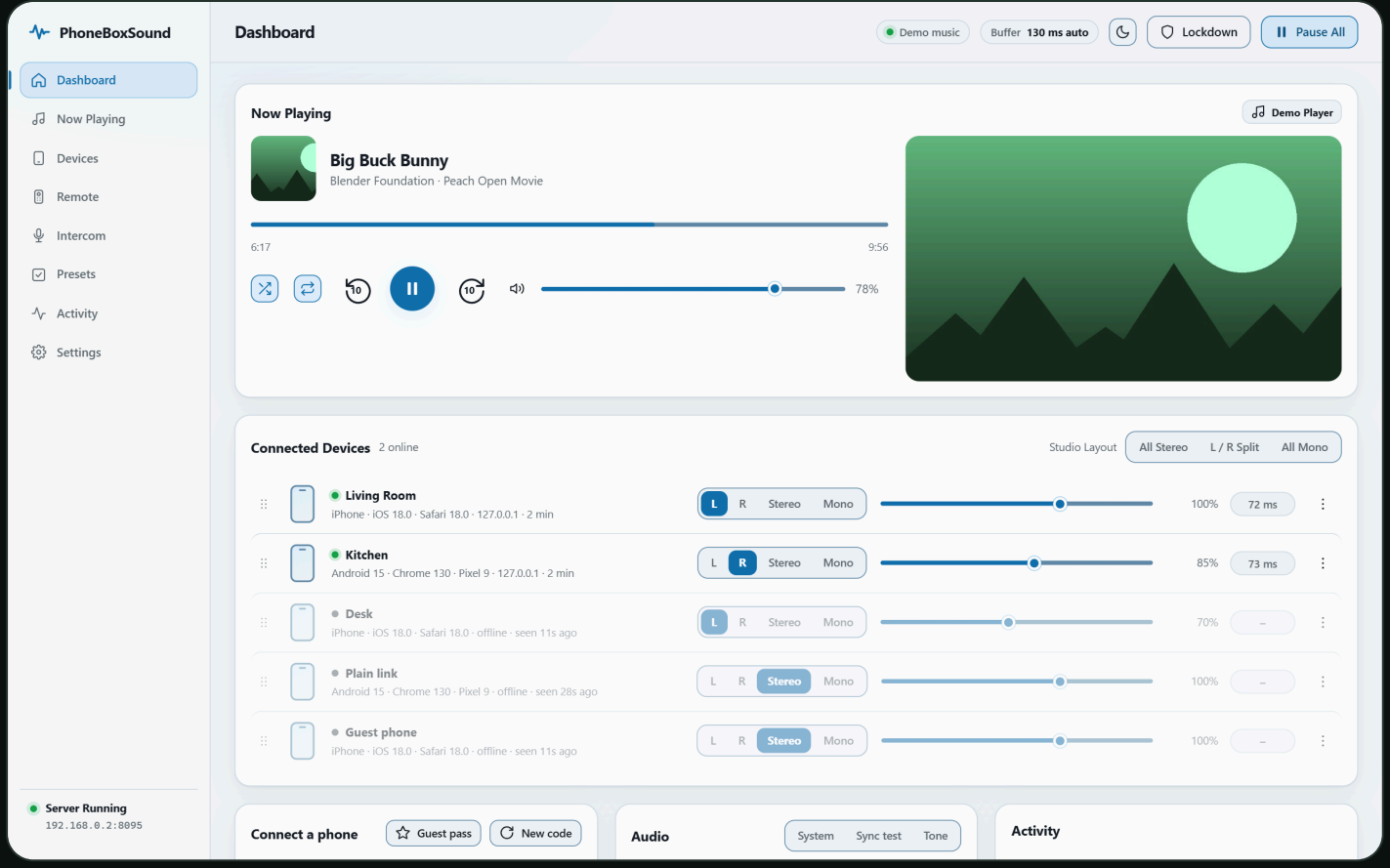
Five themes, light and dark



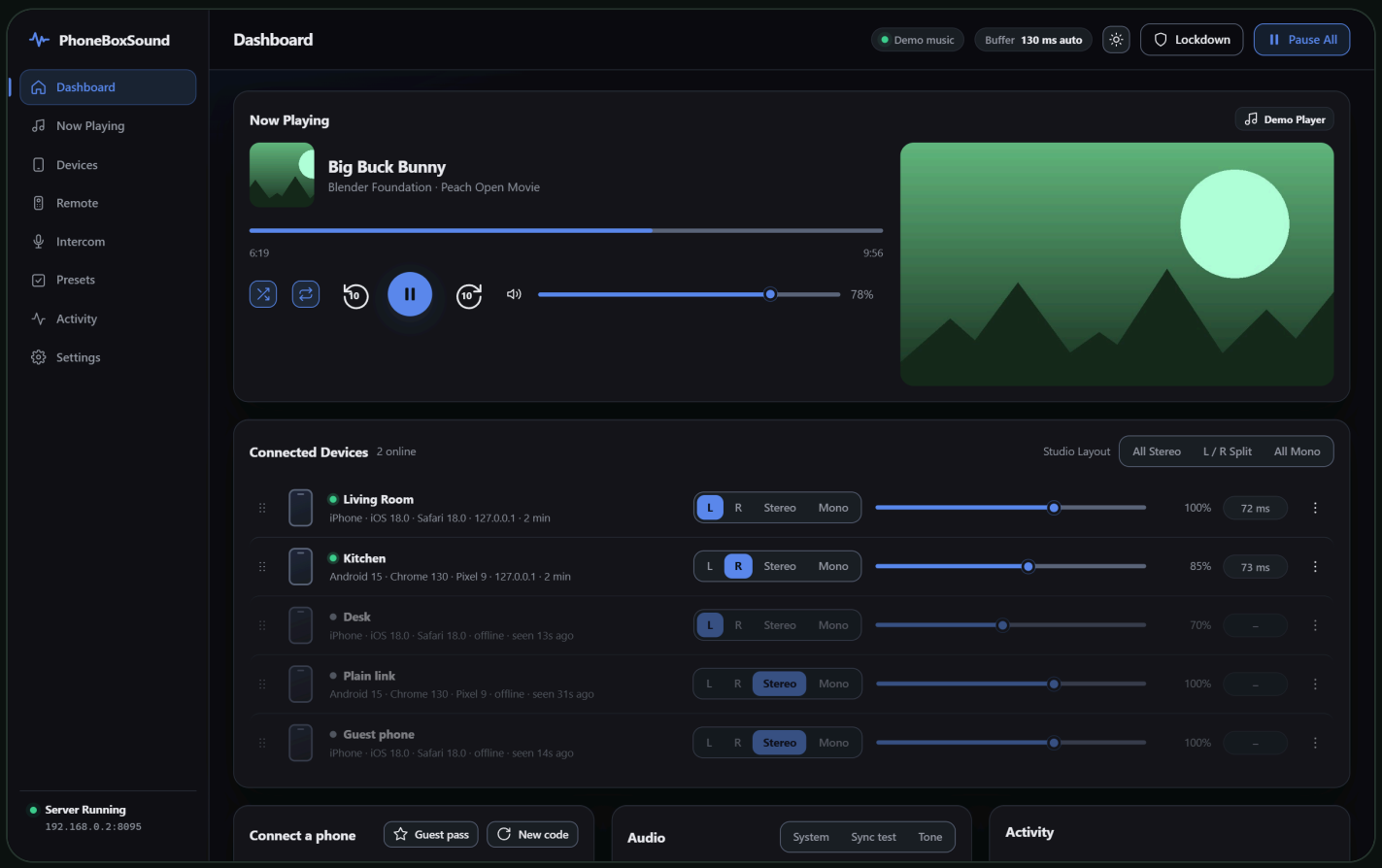
Sand · light



Aurora · dark



Ocean · light



Midnight · dark

REFERENCE

Permissions and settings

What each permission allows

PERMISSION	AT FIRST	THE PHONE CAN
Speaker	On	Plays the PC's sound on this phone, in sync with the others.
Media remote	On	Play, pause, skip, seek and volume of what is playing; see the queue; share its link.
Intercom	On	Talk to the other phones and the PC, and hear announcements.
Search and play	On	Search YouTube, Spotify and Apple Music and start a result on the PC.
Keyboard & mouse	**Off**	Move the PC's pointer, click, scroll and type: full control of whatever is on screen.
Clipboard	**Off**	Read what you copied on the PC, and put text or pictures on the PC's clipboard.
Screen view	**Off**	See the PC's screen (or the active window) live on the phone.
Power	**Off**	Lock, sleep, restart or shut down the PC, each after a confirmation and a countdown.
Manage speakers	**Off**	Change every phone's volume, channel and timing, pause all, start the bedtime timer.
Files	**Off**	Send files to the PC's share folder and download what is in it. Nothing outside that folder.
PC commands	**Off**	Press the command buttons you made in Settings. A phone can never create or change one.
Host music	**Off**	Play audio files from this phone on every phone, in place of the PC's sound.
Approve phones	**Off**	Let a waiting phone in, or turn it away, without going to the PC.
Shake to lock down	**Off**	Three firm shakes of this phone put the PC into lockdown.

Everything marked Off reaches into the PC itself. It stays off until you switch it on for that phone, is withheld from a phone that has not proved its own key while strict security is on, and from every phone during lockdown.

Every setting

SETTING	WHAT IT DOES
Start with Windows	Starts quietly when you sign in, with a tray icon. Off: it only runs when you start it.
Appearance	Dark / light / system, five themes, interface size. Phones follow unless they choose their own.
Commands	Buttons that run a program or script on this PC, for phones with the PC commands permission.
Files	The one folder phones with the Files permission can send to and download from.
Room code, addresses, port	New code any time. Changing the port moves connected phones along.
Advertise as phoneboxsound.local	Lets phones use a name instead of a number address.
Secure server / warning-free https	https on the same port. The warning-free address removes the browser warning; it needs internet.
Approve new phones	The code alone is not enough: a new phone waits until you approve it.
Lockdown	No phone controls the PC, no new phones, no remote API. Phones keep playing.
Strict security	PC controls only for phones that proved their own key on the secure address.
PC speakers in step	Delays the PC's own speakers or headphones to match the phones.
Lighter stream on poor Wi-Fi	A struggling phone gets half the data until its connection recovers, and is told to come closer.
Auto buffer, minimum, maximum	How far behind the PC the phones play. Auto adapts it to your Wi-Fi within the limits.
Lip-sync	Shifts the player's audio so phones match the picture (VLC, MPC-HC, mpv).
Player integrations	One-click VLC setup, mpv pipe, MPC full control, the browser extension.
Search keys, Spotify account	Optional YouTube key; your Spotify app's ID to link your account.

Password & encrypted storage	Dashboard password; settings encrypted with it; ask at every server start or not.
API & automation	Token, example commands, webhooks.
Diagnostics	Health at a glance and an export with no secrets in it.
Activity record (Activity page)	The two-day record of what each phone did. Switch it off entirely, or erase it.
On each phone: More	Theme, text size, timing, shake to lock down. Pad options: pointer speed, open the keyboard by itself.

DETAILS

What controls what

Players

PLAYER	SETUP	WHAT YOU GET
Chrome, Edge, Firefox, Spotify, Windows apps	None	Title, artwork, timeline, seek, play/pause, next/previous
MPC-HC / MPC-BE (K-Lite)	None	Timeline read from its window, seek, volume, speed, subtitles, fullscreen, live frame as artwork
PotPlayer	None	Timeline, seek, volume, speed, fullscreen
VLC 3	One click	Everything, plus automatic lip-sync
mpv	One flag	Everything

Safety

Pairing	The 4-digit code only reaches the waiting room: you approve each new phone. Five wrong codes lock that address out for a minute. The code can never take over a phone that is already paired.
Identity	On the secure link each phone holds a key that can't be copied off it and signs in with it. A stolen id or token is refused.
Lockdown	One button: every phone keeps playing, none can control the PC, nobody new can join.
Storage	Password-protected and encrypted. A changed, swapped or missing file stops the server from starting.
Roles	Speaker, media remote, intercom, and eight that reach into the PC itself and stay off until you enable them: keyboard & mouse, clipboard, screen view, power, managing all speakers, files, commands, hosting music. A phone can also be blocked outright.
Dashboard	Only this PC, and only with the password (or the API token). Phones can't download its code. Other websites are refused, pages can't be framed or injected into.
Network	Everything stays on your local network. Player web interfaces are bound to this PC only.

TERMS

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UNDER THE HOOD

Architecture

Server · Python

`audio.py`	WASAPI loopback capture, timestamps, packet format
`hub.py`	Devices, pairing, routing, adaptive buffer, intercom, zones, scenes
`media.py` , `players.py`	Finds what is playing and controls it
`snap.py` , `desktop.py`	Video frame for artwork; clipboard and screen view
`calibrate.py`	Chirp detection: aligns phones through the PC microphone
`app.py`	HTTP/HTTPS, WebSockets, REST API

Web app · React + TypeScript + Vite

`web/src/player`	Phone app. `engine.ts` does sync and playback outside React, so 100 packets a second never wait for a render
`web/src/control`	Dashboard; each page is a lazy-loaded chunk
`web/src/lib`	Clock sync, wire format, intercom audio, UI kit, tiny store

Builds are content-hashed, so phones always load the current version.

Run it

```
npm install          # once
npm run setup        # once: Python dependencies
npm run dev          # server + live-reloading UI  →  http://localhost:5173/control
npm run build        # production build into web/dist
npm start            # build, then run             →  http://localhost:8080/control
npm run docs         # regenerate these screenshots, videos and the PDF (also an end-to-en
npm run package      # standalone Windows build  →  release/PhoneBoxSound-win64.zip
```

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`npm run docs` from the running app.